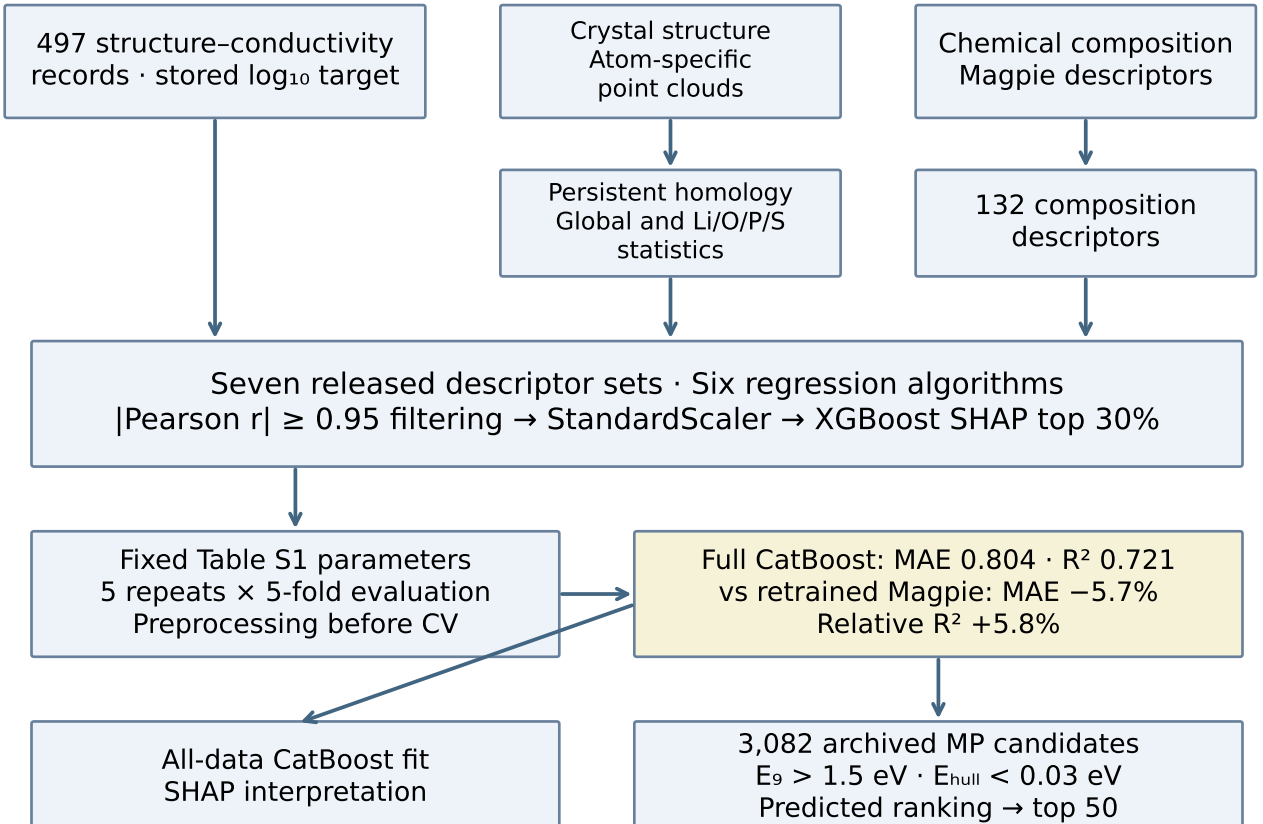
497 structure–conductivity records · stored log10 target
Crystal structure
Atom-specific point clouds
Chemical composition
Magpie descriptors
Persistent homology
Global and Li/O/P/S statistics
132 composition descriptors
Seven released descriptor sets · Six regression algorithms
|Pearson r| ≥ 0.95 filtering → StandardScaler → XGBoost SHAP top 30%
Fixed Table S1 parameters
5 repeats × 5-fold evaluation
Preprocessing before CV
Full CatBoost: MAE 0.804 · R² 0.721
vs retrained Magpie: MAE −5.7%
Relative R² +5.8%
All-data CatBoost fit
SHAP interpretation
3,082 archived MP candidates
E9 > 1.5 eV · Ehull < 0.03 eV
Predicted ranking → top 50